

Day 7 — Two's Complement in Computers

Module: Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

Goal

Show how the same idea underlies binary computer subtraction.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

Today's idea (10 min)

- *Nikhilam Navataścaramam Daśataḥ* (IAST: Nikhilam Navataścaramam Daśataḥ; lit. "all from 9, the last from 10")
- Computers use the same trick! In computers, numbers are 0s and 1s. They use "all from 1, last from 1+1" — same idea but with 1 instead of 9.

Show a calculator. *"When you press 100 - 37, the computer secretly uses our trick to find the answer fast."*

Children share: *"What's the coolest computer thing they know?"*

Activity (8 min)

A simplified version of *Complement Flashcards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.

- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.